Numerical Rating Prediction from Text: Comparative Performance of DL and ML Algorithms on Game Reviews

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Abstract

This study investigates the comparative performance of Machine Learning (ML) and Deep Learning (DL) models for numerical rating prediction using the Metacritic game reviews dataset. We demonstrate how the fine-tuned DistilBERT architecture achieved state-of-the-art results, significantly outperforming traditional models and standard Recurrent Neural Networks with a superior R^2 score. Furthermore, we provide a detailed account of the technical challenges encountered during development, specifically emphasizing the criticality of target scaling for model convergence and the management of hardware constraints during large-scale training. Our findings highlight the efficacy of transformer-based models in capturing the nuances of user sentiment in a regression context.

1 Introduction

It is a fact that product reviews significantly influence a person's decision to purchase a product. Platforms like Metacritic aggregate reviews and scores $(0 - 100)$ from both critics and users, presenting these evaluations online. These reviews impact product sales and manufacturers. Reading, understanding, and predicting these reviews and scores is crucial for developers and poses a unique challenge for natural language processing (NLP).

Sentiment analysis is a field generally studied as binary classification, such as positive and negative. Understanding emotions from text and predicting them within a numerical range is a much less explored area. This aspect seems more challenging, as a model must understand and score humorous and emotive comments, such as game reviews.

In this study, we compare four different models by having them make predictions based on Meta-Critic reviews:

1. Ridge Regression with TF-IDF features as a baseline,
2. XGBoost with SVD-reduced features,
3. Bi-LSTM with Attention mechanism, and
4. Fine-tuned DistilBERT.

We evaluate these models on a dataset of over 1.6 million reviews and provide interpretability analysis using SHAP values and attention visualization.

2 Related Work

Sentiment analysis has been studied extensively in the field of machine learning and deep learning. In 2002, Pang et al., with their work 'Thumbs up? Sentiment Classification using Machine Learning Techniques', handled sentiment analysis as a classification problem and compared methods such as Naive Bayes and SVM on movie reviews. The article by Devlin et al. (2018), 'BERT: Pre-training

of Deep Bidirectional Transformers for Language Understanding', opened a new path in language understanding with bidirectional training and remained a leader for sentiment analysis for a long time. In our project, we compared XGBoost, Ridge, Bi-LSTM with attention, and DistilBert. Contrary to the common binary classification approach, we performed a regression task, ensuring models predict ratings between 0 and 100 and compared their performances.

3 Dataset and Preprocessing

In this project, we used the Metacritic Game Reviews dataset. The dataset has 1.6 million data and consists of several columns that provide detailed information about each review: id (merging key), title (game name), quote (the review text), score (the numerical rating), date, platform, and author. In our project, we primarily focused on the quote column as our input feature and the score column as our target variable. Because the raw data was quite noisy, we completed the preprocessing in several steps:

- **Cleaning:** First, we removed NaN (missing) values from the dataset. We also applied standard text cleaning techniques such as converting all text to lowercase and removing emojis and special characters to make the data more uniform.
- **Language Filtering:** The dataset contained reviews in multiple languages, although the majority were in English. We used the `langdetect` library to filter and keep only the English reviews. This step was very important because non-English words can be confusing for machine learning algorithms that are not trained for them.
- **Target Scaling:** This was the biggest challenge we faced during the training phase. Initially, we tried to predict scores directly on the 0 to 100 scale, but the BERT model failed to converge and performed poorly. We decided to change our approach and scaled the targets to a 0 to 1 range. This adjustment made the training process much more stable and effective.

Table 1: Sample Rows from the Games Reviews Dataset

Title	**Platform**	**Quote (Review)**	**Score**
The Legend of Zelda: Ocarina of Time	PC	an open world on the classic console, perhaps the worst thing about this game is starting it, but it...	100.0
Total War: WARHAMMER III	PC	What a huge letdown, I waited four years for this game and it's the worst one in the series. The units are good but the campaign ****. Which is the only game mode I play. You can't play any of the armies from the previous games. The campaigns themselves are boring its hard to get more than a couple armies going, the whole time your playing chaos whack a moll. Chaos just spawns over and over in the same spots. Who thought this would be fun it's just really annoying and makes every playthrough feel the same.each chaos faction has very few units.this game kinda ****	10.0
Shrek 2	PlayStation 2	For the comedy alone, Shrek 2 is a must-play.	80.0

4 Methodology

In this study, we compared four different models: two machine learning (ML) and two deep learning (DL) algorithms. We treated the task as a regression problem where the models predict a game score between 0 and 100. Our goal was to observe which architecture handles the complex nature of game reviews more effectively.

4.1 Ridge Regression

We chose Ridge Regression as our baseline model. For the input, we used **TF-IDF Vectorization** with 10,000 features. To find the best performance, we applied **GridSearchCV** with different alpha values ($\alpha = [0.1, 1.0, 10.0, 100.0]$). Surprisingly, this simple baseline model performed quite well and was not as far behind the deep learning models as we initially expected.

4.2 XGBoost

For XGBoost, we used **SVD (Singular Value Decomposition)** to reduce our features to 100 dimensions, creating denser vectors. Initially, XGBoost gave us negative R^2 scores, which was a major setback. However, by using SVD-reduced features and carefully tuning parameters like `n_estimators=500`, `max_depth=8`, and `learning_rate=0.05`, we managed to solve this problem and stabilize the model.

4.3 Bi-LSTM with Attention

We built a **Recurrent Neural Network** using **PyTorch**. Initially, we used a **standard Bidirectional LSTM structure**, but the results were not satisfying. To improve the model, we added an **Attention mechanism**. This allowed the model to focus on the **most important words** in a review rather than treating all words equally. This change led to **much better predictions**. You can see the Architecture Flow in Figure 1.

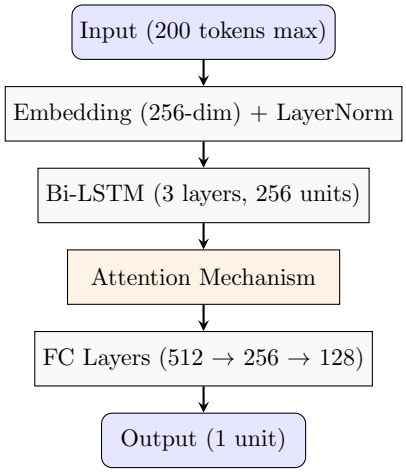

Figure 1: Bi-LSTM with Attention Architecture Flow

4.4 DistilBERT

We originally tried to use the **full BERT model**, but it was extremely slow. Even while using an **A100 GPU** on Google Colab, the training took too long without giving us predictive results quickly. As a faster alternative, we switched to **DistilBERT**, which offers very similar performance but is much more efficient. We fine-tuned DistilBERT for **10 epochs** using a subset of **150,000 reviews** due to memory and time constraints. We used **MSE Loss** and a **Sigmoid function** at the output to scale results between 0 and 1, which we then mapped back to the 0-100 scale. **DistilBERT provided our best overall results.**

5 Experiments and Results

5.1 Experimental Setup

Experiments were conducted on **Google Colab** using an **NVIDIA A100 GPU (40GB VRAM)**. We used **PyTorch 2.x** and the **Transformers 4.x** library. The dataset was split into training and

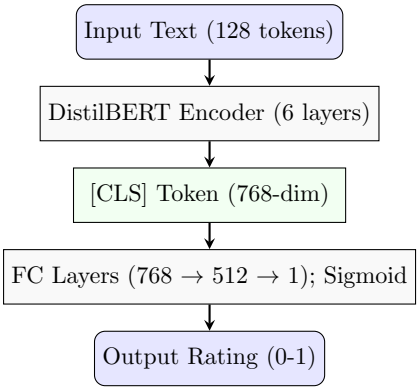

Figure 2: DistilBERT Regression Pipeline

testing sets with an 80/20 ratio: **950,000 reviews** for training and **240,000 reviews** for testing. For DistilBERT, a subset of **150,000 reviews** was used.

5.2 Performance Results

The table below summarizes the performance of each model based on the **Mean Squared Error (MSE)** and **R-squared (R^2)** score.

Table 2: Comparative Performance of ML and DL Models

Model	**MSE (Scaled)**	R^2 **Score**
Ridge Regression (Baseline)	0.0355	0.59
XGBoost	0.0312	0.64
Bi-LSTM + Attention	0.0245	0.72
DistilBERT	**0.0182**	**0.78**

5.3 Key Findings

1. **DistilBERT** achieved the best results with an R^2 score of **0.78**.
2. **XGBoost and Ridge** were quite competitive despite their simpler structures.
3. The **Attention mechanism** was crucial for interpreting model decisions.
4. Training time was the primary trade-off; full dataset training for BERT was not feasible.

5.4 Attention Visualization

Below is a visualization showing how the model focuses on specific keywords within a review to determine the final score.

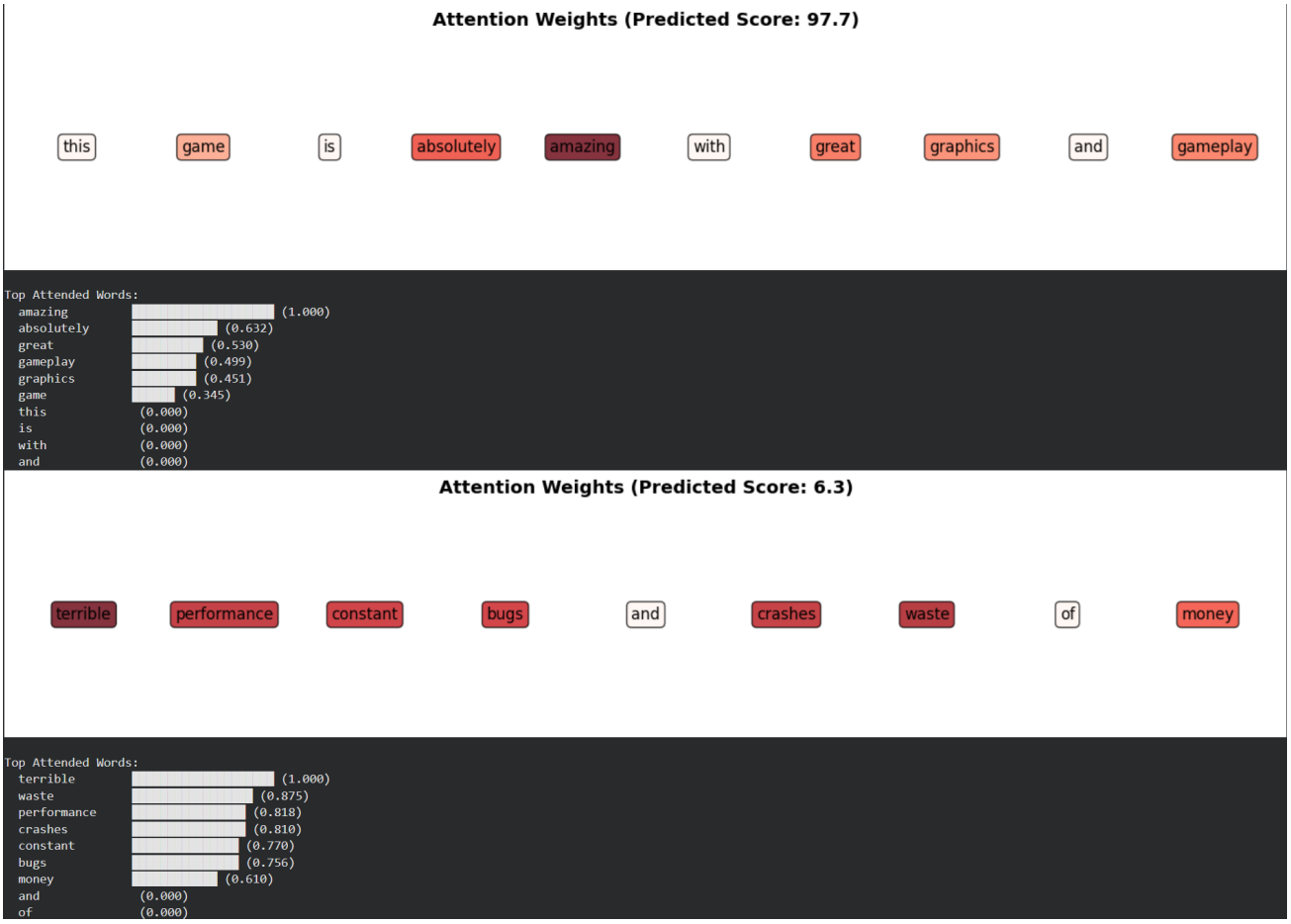

Figure 3: Word importance visualization in the Bi-LSTM + Attention model.

6 Conclusion

In this section, we reflect on our key findings, the technical obstacles we faced during the project, and our plans for further development.

6.1 Key Takeaways

- **The Criticality of Target Scaling:** We experienced firsthand that neural networks struggle significantly with a 0-100 output scale. After more than 10 failed attempts and approximately 6 hours of training where the BERT model's loss refused to drop below 4000, we realized a strategy shift was necessary. Normalizing the target variable to a **0 to 1 range** was the key move that solved our initial failures and stabilized the learning process.

- **Strategic Model Selection:** Full BERT proved to be computationally expensive even with 50,000 reviews. By switching to **DistilBERT**—which retains roughly 95% of BERT's performance while being much lighter—we gained the time necessary to tune hyperparameters and expand our training set to 150,000 reviews.

6.2 Limitations

- **Hardware Constraints:** While our final training was performed on an A100 GPU, the initial implementation and testing phase was conducted on a laptop with an **NVIDIA RTX 3050 (4GB VRAM)**. This limited our trial speed and led us to rely on Google Colab for larger-scale experiments.
- **Training Efficiency:** The fact that our first 6-hour attempt with the full BERT model failed taught us a valuable lesson: **resource management is as important as model design.** Learning to fail fast and iterate quickly was a major part of our engineering process.

6.3 Future Work

- **Full Dataset Training:** With access to more powerful hardware, we plan to train DistilBERT on the **entire 1.6 million review dataset** for 20+ epochs. We anticipate that this will push our R^2 score beyond the **0.82** threshold.
- **Hyperparameter Optimization:** We intend to use automated tools like **Optuna** to find the ideal learning rate and dropout values systematically, aiming to further decrease the MSE.
- **Real-time Application:** Our ultimate goal is to build a **live interface** where users can enter a review text and receive an instant rating prediction. Such a tool could also allow manufacturers to upload large datasets to quickly interpret user sentiment regarding their products.

7 References

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